

The Lab 2026

Brushless Car Wash - 3D Reconstruction

The Three

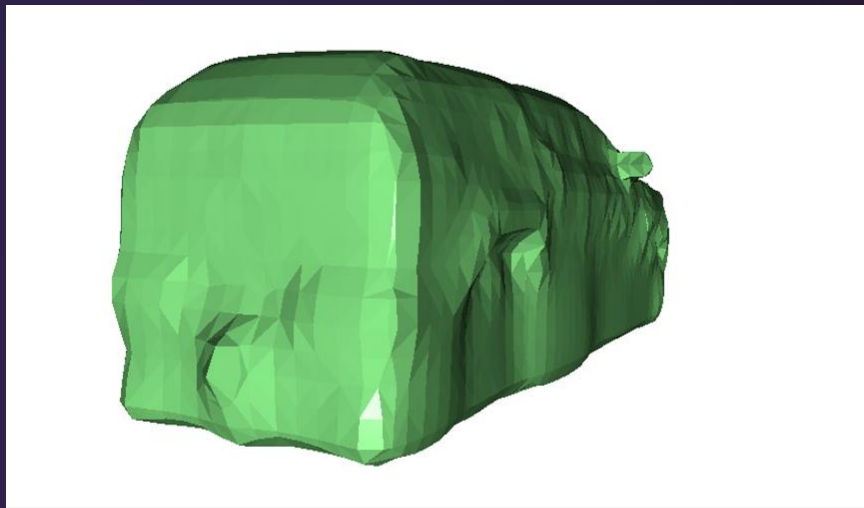
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Before & After

Visual comparison of the raw scan vs. processed output



Scanning Car



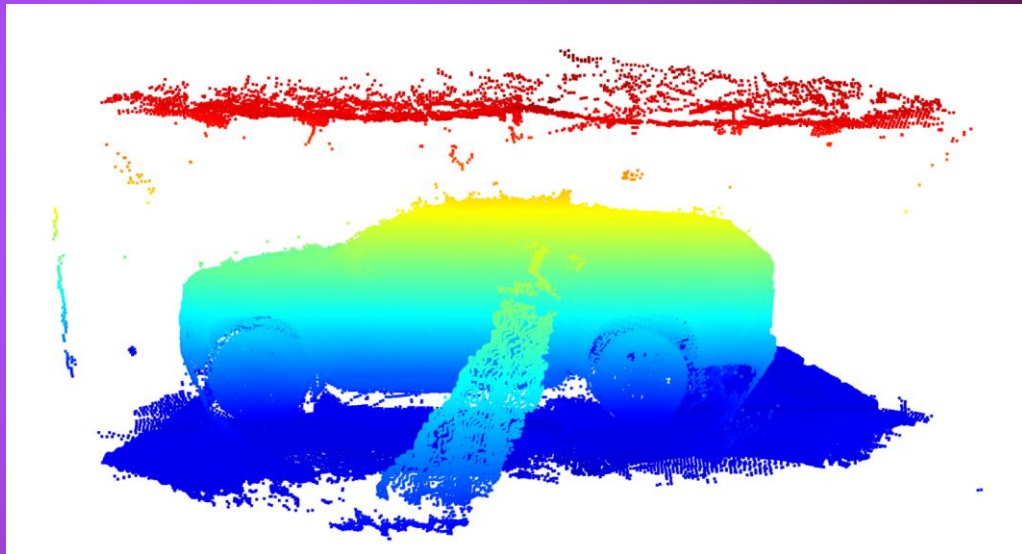
Clean Mesh

The First Step

We want to get rid of unwanted data

What counts as unwanted data?

- ▶ Floor
- ▶ Ceiling
- ▶ Isolated clusters
- ▶ Walls
- ▶ Statistical outliers



Project Goal

Transform raw .pcd point cloud data into a precise, watertight 3D mesh (STL)

Key Requirements

Preserve critical vehicle features:

- ▶ Mirrors
- ▶ Wheel areas
- ▶ Tow Bar
- ▶ Bull Bar
- ▶ Roof
- ▶ Main car body

Final Objective

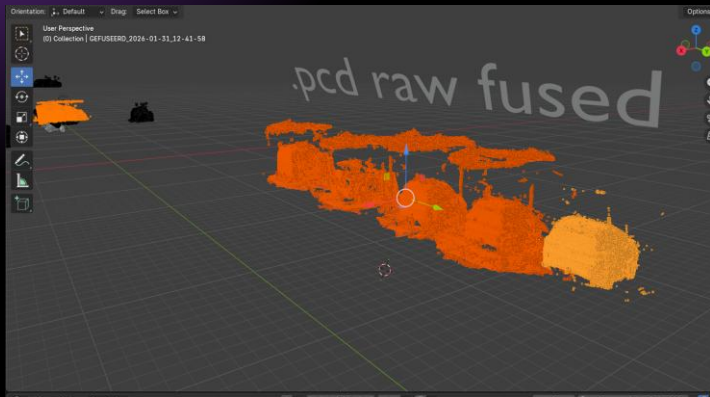
Deliver a production-ready mesh:

An accurate 3D mesh that ensures safe, brushless car washing operations.

- ▶ No collision with vehicle features
- ▶ Watertight STL export
- ▶ Production-ready quality

Work Plan

End-to-end pipeline from raw scan to watertight mesh



1. Extract raw point cloud data
2. Remove all detected noise
3. Detect important car features
4. Handle abnormal or unsafe data
5. Convert to watertight 3D mesh
6. Export .stl

Tech Stack

Python

Libraries:

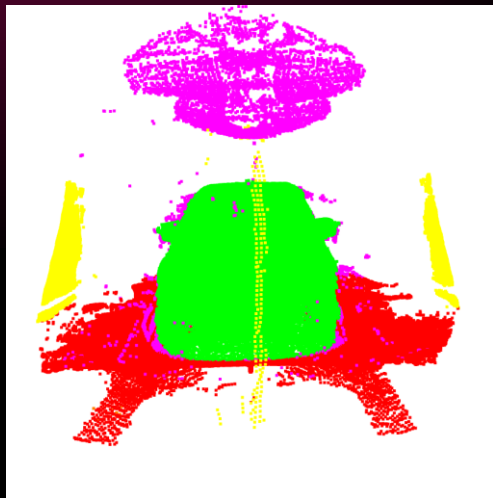
- Open3D
- PyMeshfix

Tried:

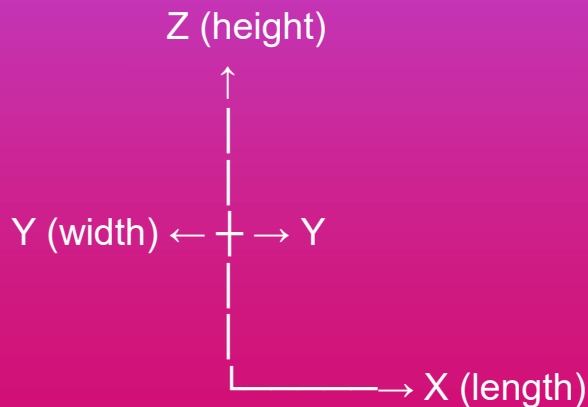
- TryMesh
- ShapeLi
- Meshlib

Viz: Blender,
Open3D

Filtering mechanism



To isolate the car from the raw point cloud, we implemented a multi-stage filtering pipeline combining geometric detection and statistical analysis



Explored Strategies

Meshing

Alpha Shape reconstruction

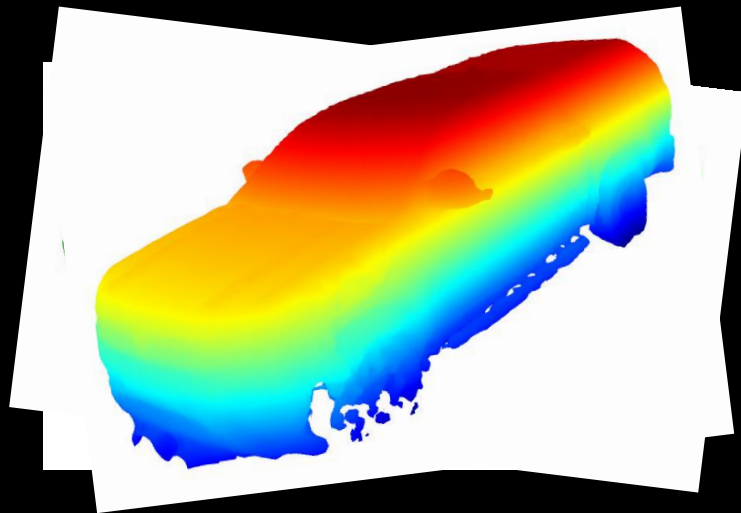
“Cover” the car

Poisson reconstruction

Voxel + normal (vector) math

Ball Pivoting reconstruction

Roll a “ball” over the car



Smoothing

Average Filter smoothing

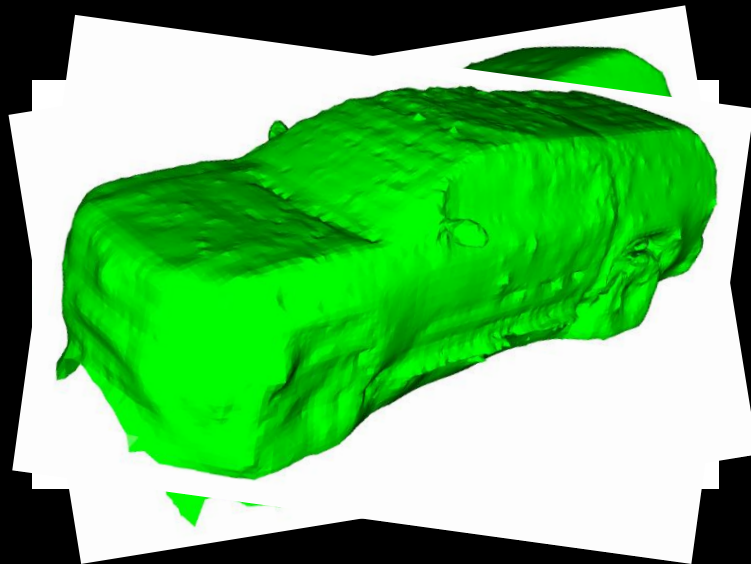
Average vertices

Downsampling smoothing

Downsample before meshing
"Averaging points"

Laplacian Filter smoothing

Vertices with weights



Keeping Details

Detecting Sides with Planes

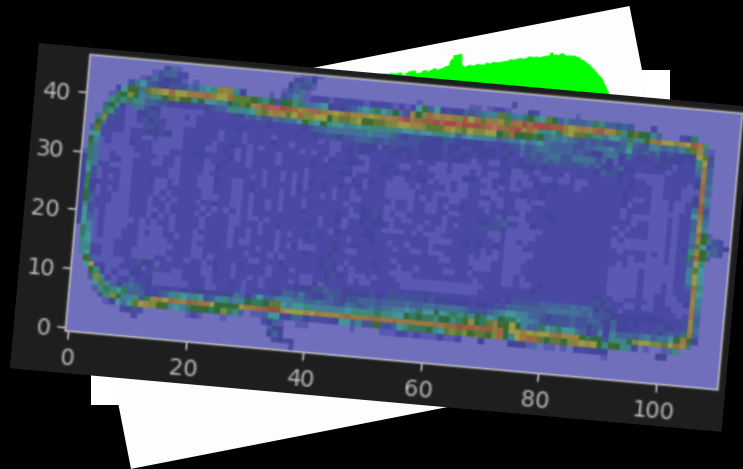
Extract estimated sides of car by fitting planes to them and cutting at the plane

Detecting Sides with Density

Extract estimated sides of car by density on z plane of the pointcloud and cutting at those x/y coordinates

Detecting Sides with Coordinates

Extract estimated sides of car by distance from side of the pointcloud and cutting at that distance



Smooth "Curtain"

Curtain with Cutting

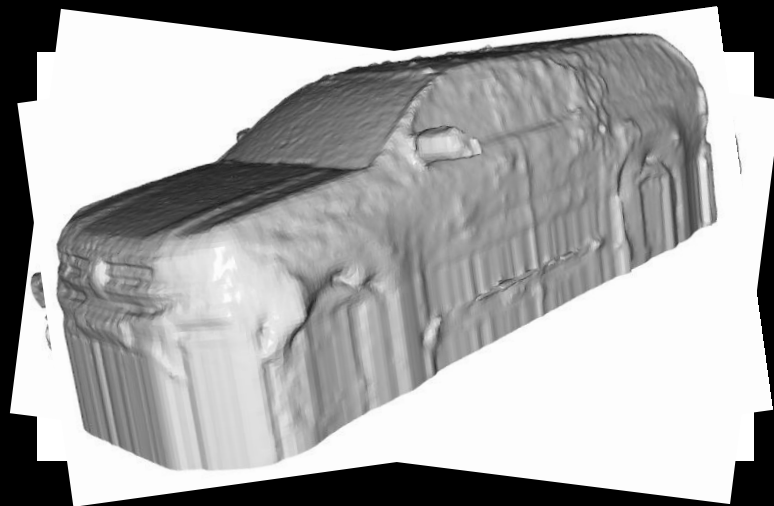
Cut mesh at calculated height
Pull curtain down from cut car

Curtain without Cutting

Pull curtain down from calculated height

Curtain with Interior Removal

Pull a curtain from calculated height
Remove "hidden" points



Challenges

Challenges

Normals

- 1. Normals were pointing upwards on the sides
- 1. Normals were pointing inwards in some places
- 1. Paid library worked
- 1. Free library worked after more experiments

Smooth but Detailed

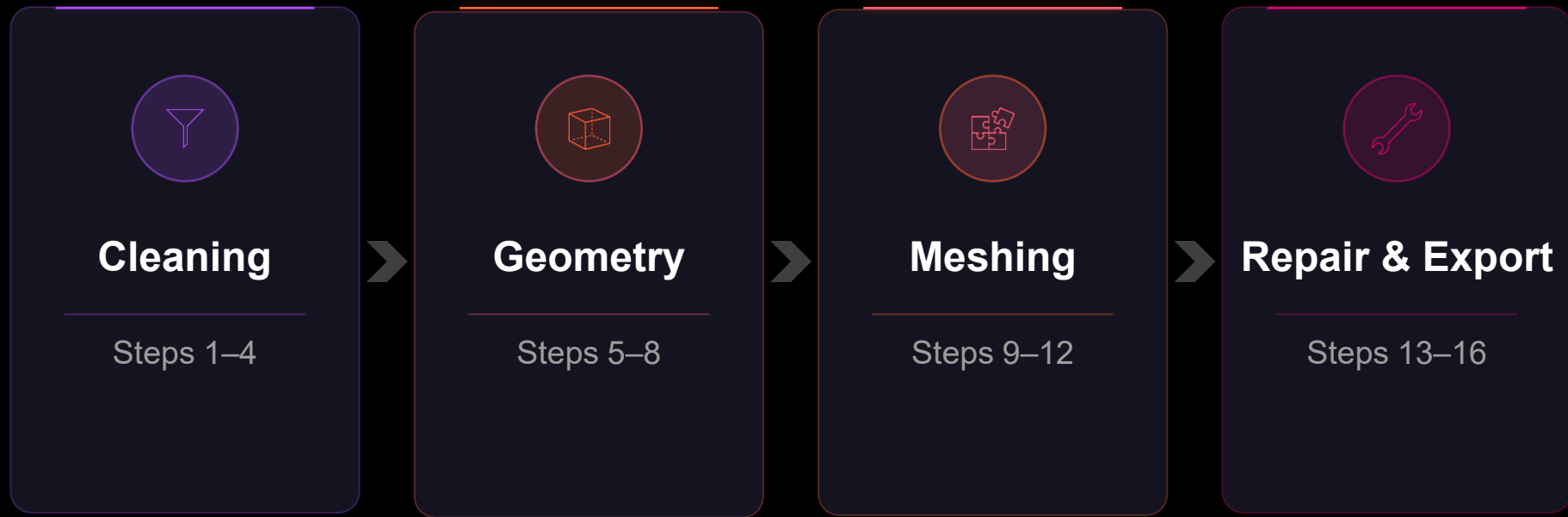
- 1. Too much detail (and noise)
- 1. Loss in detail
- 1. Fine Balance

Pipeline Architecture

14 steps from raw point cloud to watertight mesh

Pipeline Phases

16-step automated pipeline from raw scan to export-ready model



Noise Removal

Steps 1–4 Removing unwanted noise from raw scan data

01 Downsample

Reduce point density while preserving key features

02 RANSAC Filtering

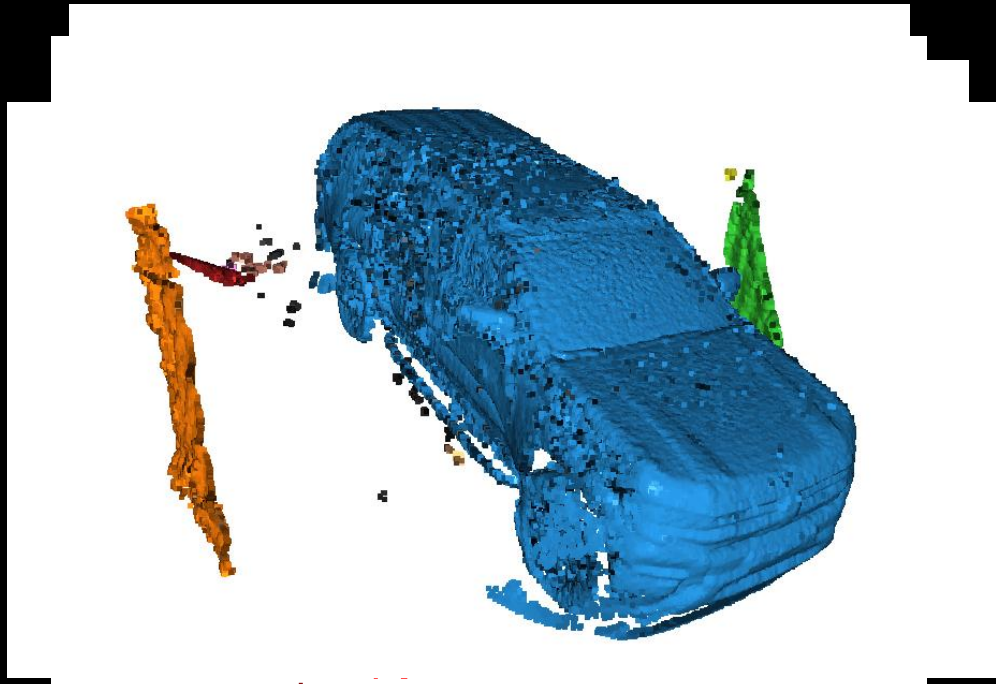
Identify and remove planes and floor/height filter

03 Statistical Outlier Removal

Eliminate points deviating from local distributions

04 DBSCAN Clusters

Identifies cluster and removes all other than car



Recovery & Correction

Steps 5–8 Restoring and correcting scan geometry

05 Raw Recovery

Reconstruct any missing data from overlapping scans

06 Identify regions and thinning

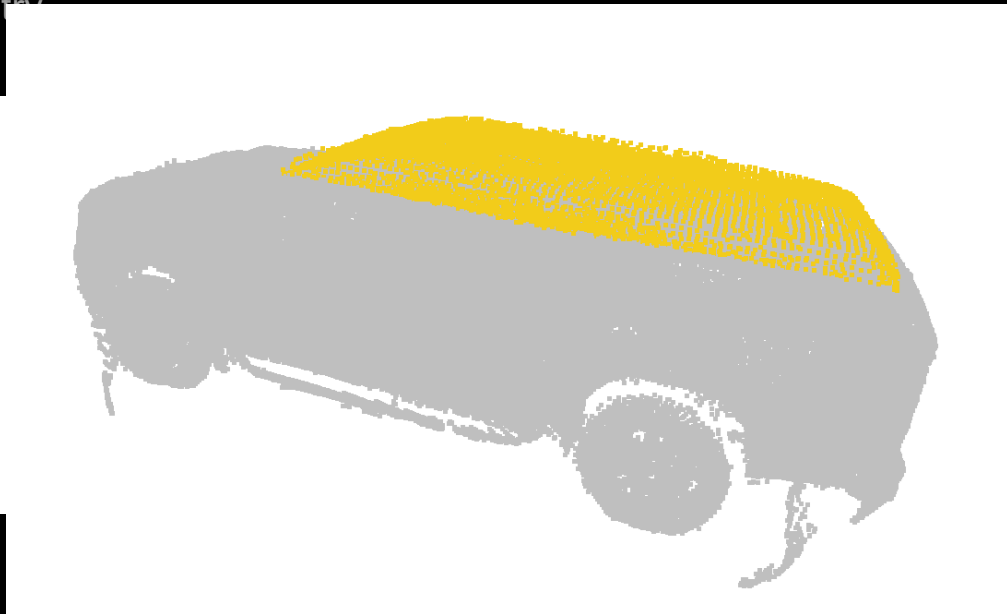
Regions such as front/rear, side, roof, and mirrors

07 Roof spikes

Areas above car roof are removed

08 Identifying roof and filling

Once the roof is found and there are gaps, it is filled



Final Cleanup & Validation

Steps 9–11 Final refinements and quality checks

09 Identify wheel curves/edges

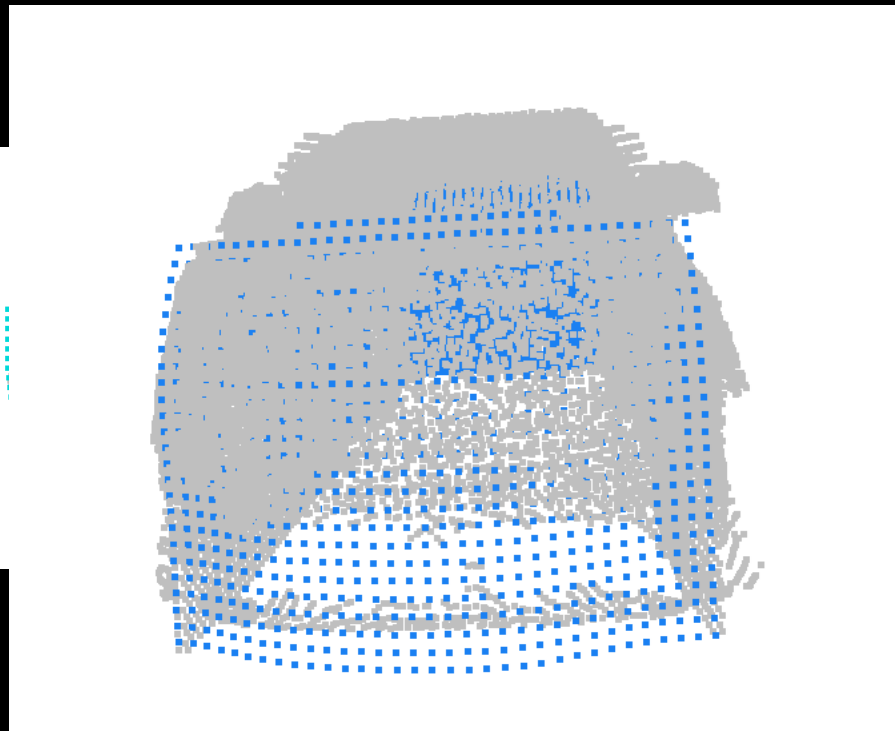
Sharpen boundary edges and clean up transitions

10 Identify mirror

Detect and eliminate remaining visual artifacts

11 Place fill for walls and bumper

Automated QA checks before passing to meshing



Feature Detection

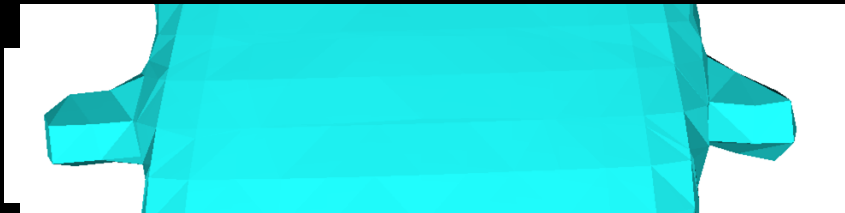
Steps 1–3 Identifying key geometric features in the point cloud

12 Poisson Reconstruction

Connects the dots

13 Mirror snapping

Adjusts mirrors to match raw closer

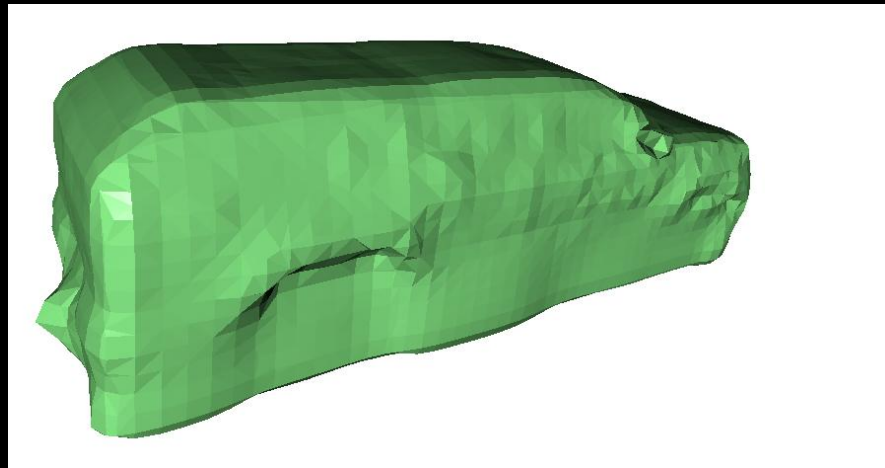


Repair & Export / Watertight Check

Watertight Check

Final validation

14 Final check, water tightness



Future Work

- 3D Deep learning models (PointNet, PointNet++)

Thank you!

Check out our website!

